TBRNet: Two-Branch Reinforcement Network for Few-Shot Semantic Segmentation in Remote Sensing Images

Xin Wang, Yi Jiang, and Huiyu Zhou

***Abstract*—Traditional semantic segmentation methods for remote sensing images (RSIs) require abundant labeled data yet falter when samples are scarce. Few-shot semantic segmentation (FSS) innovatively resolves this data scarcity bottleneck. However, existing FSS models face unique challenges in RSIs: natural-to-remote-sensing domain gaps, intraclass variance, and multicategory coexistence-induced generalization collapse. These problems may lead to the loss of discriminative query features in model performance. In this article, we present a novel two-branch reinforcement network called TBRNet to tackle these challenges. Specifically, we first propose a prototype reinforcement module (PRM) to generate enhanced context-adaptive prototypes by dynamically weighting query-support feature contexts, which effectively mitigates intraclass variance via strengthening support images' perception of discriminative query features. In addition, to deal with the challenge of the coexistence of multiple target categories, we develop a multilevel guidance reinforcement module (MGRM), which provides multilevel guidance maps across resolutions to model cross-level semantic dependencies and emphasize discriminative subregions. Extensive experiments conducted on the iSAID-5^i and DLRSD-5^i remote sensing (RS) datasets have shown the superiority of our proposed TBRNet compared with several state-of-the-art approaches. Ablation studies have also verified the effectiveness of the proposed modules. Our source code is made publicly available at https://github.com/WangXin81/TBRNet**

***Index Terms*—Few-shot learning, few-shot semantic segmentation (FSS), prototype, remote sensing images (RSIs).**

I. Introduction

SEMANTIC segmentation, a fundamental task in remote sensing (RS) [1], [2], [3], aims to assign pixel-level labels to highlight the most meaningful semantic content in an image. It plays a pivotal role in diverse RS applications, including urban planning [4], [5], [6], agricultural monitoring [7], [8], [9], [10], and other areas [11], [12], [13], [14], [15]. In recent years, deep learning (DL)-based algorithms [16], [17], [18] have dominated the field, leveraging their superior ability to extract and represent discriminative features [19], [20], [21].

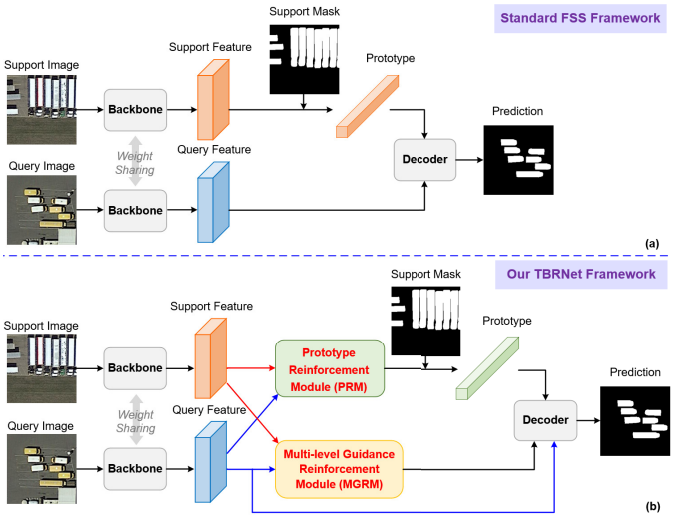


Fig. 1. Comparison between (a) standard FSS framework and (b) our TBRNet FSS framework for RS semantic segmentation.

Despite their success, conventional deep neural networks (DNNs) for RS segmentation face a critical bottleneck: vast amounts of accurately labeled data [22]. The high resolution of RS images exacerbates this by drastically increasing annotation pixels, making pixel-level labeling extremely resource-intensive. For example, manually labeling millions of pixels in complex urban scenes—filled with intricate structures—demands significant time and cost [23]. To address these two major obstacles—high-quality data scarcity and annotation costs—few-shot semantic segmentation (FSS) has emerged as a solution. As a paradigm, FSS enables effective model adaptation with minimal labels [24], [25], transferring prior knowledge from limited support images to overcome fully supervised limitations in scarce-data scenarios [26].

Since its emergence, numerous FSS approaches have refined the prototype-based strategy [27], [28], [29]. A common approach is to extract category-specific prototypes from support images and align query image features with these prototypes for segmentation, as illustrated in Fig. 1(a). For example, PANet [30] computes pixelwise distances between query and support features to generate initial segmentation

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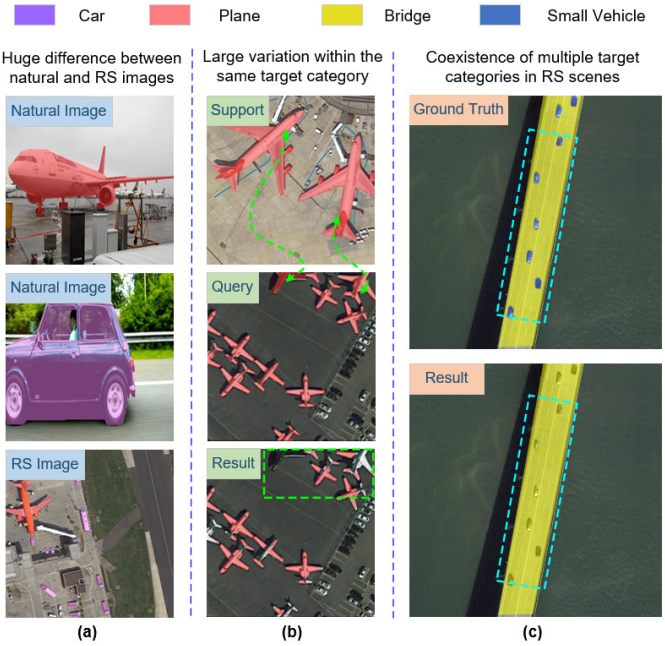


Fig. 2. Challenges of FSS in RSIs. (a) Huge difference between natural and RS images limits the performance of current FSS models. (b) Large variation within the same target category leads to false predictions. The green lines indicate the target differences between the support and query images. (c) Coexistence of multiple target categories causes a generalization breakdown. The cyan boxes indicate the coexistence of different target categories.

masks, which are then refined for query image segmentation. CANet [31] advances this by using a convolutional prototype linking to model spatial relationships, combined with atrous spatial pyramid pooling (ASPP) [32] to capture multiscale context. PFENet [33] introduces a prior knowledge module that fuses prototypes, masks, and query features through a hierarchical fusion mechanism, enhancing cross-scale feature consistency.

While effective in natural images, these methods fall short in RS due to three unique challenges (as illustrated in Fig. 2 with samples from the well-known iSAID-5^i dataset [34]).

1) *Domain Gap Between Natural and RS Images:* RS images are captured from a top-down perspective over long distances, leading to loss of fine-grained details [e.g., wheels of cars in Fig. 2(a)], which undermines the discriminative power of features used in natural-image FSS.
2) *Severe Intraclass Variance:* RS objects exhibit diverse appearances due to sensor differences, seasonal/weather variations, and geographical contexts. For instance, the category "plane" in support and query images may differ significantly in size, shape, or background [see Fig. 2(b)], causing prototype mismatch.
3) *Multicategory Coexistence:* RS scenes are highly complex, with multiple classes [e.g., "bridge" and "small vehicle" in Fig. 2(c)]. Standard FSS methods often struggle to distinguish these coexisting categories, leading to generalization collapse.

To address these RS-specific challenges, we propose a two-branch reinforcement network (TBRNet), a TBRNet comprising a prototype branch and a guidance branch, which together enhance feature discriminability and contextual reasoning [see Fig. 1(b)].

First, we propose a prototype reinforcement module (PRM) for intraclass variance on the prototype branch. Standard FSS frameworks [see Fig. 1(a)] generate class prototypes exclusively from support images, ignoring the contextual differences between support and query images [35], [36], [37]. This single-source generation leads to discriminative feature loss when intraclass variations (e.g., appearance differences due to imaging conditions) arise. In contrast, PRM introduces a bidirectional interaction mechanism that bridges support and query images to generate context-adaptive prototypes.

1) *Query Feature Customization (QFC):* Unlike existing methods relying solely on support features, PRM first employs a bidirectional interaction mechanism to extract context-relevant regions from the query image, guided by support-image semantics. This "QFC" highlights pixels in the query image that are semantically aligned with the support class [e.g., identifying consistent structural patterns of "plane" across different resolutions in Fig. 2(b)].
2) *Activation and Refactoring:* The customized query features are then activated and integrated with support features to refactor support features and generate enhanced prototypes. By dynamically weighting query-support feature interactions, PRM adapts support prototypes to the specific context of the query image, mitigating the negative impact of intraclass variance. For example, when the query image shows a "plane" with partial occlusion, PRM strengthens the prototype's sensitivity to preserved structural features (e.g., wings), improving segmentation robustness.

Second, we develop a multilevel guidance reinforcement module (MGRM) for multicategory coexistence on the guidance branch. Existing guidance-based FSS methods generate guidance maps at a single feature resolution [33], [38], a limitation that hinders capturing hierarchical semantic dependencies in complex multicategory scenes where multiple categories coexist. MGRM addresses this by introducing a hierarchical guidance mechanism.

1) *Hierarchical Guidance Extraction:* Unlike traditional single-level approaches, MGRM generates guidance maps across multiple levels. Starting with a high-resolution guidance map to retain rich semantic information [e.g., activating "bridge" areas in Fig. 2(c)], each subsequent level downscales and refines the map to emphasize discriminative subregions while preserving critical details, avoiding the detail loss inherent in single-resolution methods.
2) *Cross-Level Dependency Modeling:* MGRM establishes a cross-level dependency chain. High-resolution maps guide the refinement of low-resolution maps to convert coarse semantic activations into fine-grained distinctions, while low-resolution maps supplement high-resolution maps with global contextual information, enabling the latter to prioritize contextually relevant

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regions, forming a hierarchical dependency chain that balances detail preservation and semantic consistency [e.g., ensuring "car" is distinguished from "bridge" in Fig. 2(c)].

Finally, the outputs of PRM (enhanced prototypes) and MGRM (multilevel guidance maps) are fed into a hierarchical decoder that adaptively integrates these cues with query features. Unlike traditional decoders, which often suffer from feature misalignment and semantic ambiguity, our decoder hierarchically fuses diverse cues, ensuring consistent cross-scale and cross-category alignment for precise segmentation. Experiments on iSAID-5^i and DLRSD-5^i demonstrate that TBRNet outperforms state-of-the-art methods, with significant advantages in scenarios where existing strategies fail due to severe intraclass variance or complex category coexistence.

Our main contributions are given as follows.

1) A novel framework named TBRNet is proposed to address three critical challenges in RS FSS: domain gap, intraclass variance, and multicategory coexistence.
2) PRM is developed to tackle intraclass variance. It bidirectionally fuses support and query image contexts to generate context-adaptive prototypes, reducing the negative impact of intraclass variance on segmentation robustness.
3) MGRM is introduced for complex category coexistence. It generates multilevel guidance maps across resolutions to model cross-level semantic dependencies, improving discriminability between coexisting categories in intricate scenes.
4) A hierarchical decoder with adaptive feature integration is designed to resolve feature misalignment and semantic ambiguity in traditional decoders, further enhancing segmentation performance in RS images (RSIs).

The rest of this article is structured as follows. Section II provides a brief review of the related work. In Section III, the proposed TBRNet is introduced in detail. In Section IV, experimental results along with a comprehensive analysis are presented. Conclusions are drawn in Section V.

II. Related Work

A. Semantic Segmentation

The concept of semantic segmentation was initially put forward by researchers in the 1980s. Recently, the progress of DL has greatly promoted its development.

The fully convolutional network (FCN) [22] is a milestone in the formal application of convolutional neural networks (CNNs) to semantic segmentation. In FCN, fully connected layers are replaced by convolutional layers to enhance segmentation accuracy. Later, to address the issue of information loss in FCN, DeepLab [39] was proposed. It expands the receptive field of the convolution kernel and fuses multiscale features through atrous convolution and ASPP [40], [41]. To improve upon the framework of FCN, U-Net [42] with an encoder–decoder structure and skip connections was introduced. The encoder is used to obtain image features, while the decoder is responsible for generating the final segmentation results. SegNet [43] also employs an encoder–decoder architecture and upsamples the output of the encoder according to the indices saved during the pooling step. RefineNet [44] integrates feature maps with different resolutions and gradually upsamples the maps to improve the understanding of contextual information. The pyramid scene parsing network (PSPNet) [45] proposes establishing long-distance dependencies among pixels and capturing the global information of the image.

Although the above methods have achieved remarkable performance in semantic segmentation, their reliance on supervised learning necessitates a substantial amount of labeled data for training. In RSIs, however, acquiring high-quality annotations is particularly costly and time-intensive. Moreover, these methods struggle with novel categories, leading to problems such as misclassification and unreliable predictions from poor generalization. The aforementioned issues urgently need to be resolved.

B. Few-Shot Semantic Segmentation

FSS, an effective approach for addressing the shortage of labeled data, can achieve semantic segmentation with a minimal number of labeled samples. OSLSM [46] is a classic FSS method, in which a conditioning branch is designed to enhance the FCN framework. As a conditional network, co-FCN [47] extracts different features from support and query images for segmentation. SG-One [48] designs a masked average pooling (MAP) strategy to compute support prototypes, which has been widely used in subsequent models.

As various methods progress, the mainstream concept of FSS evolves into integrating prototypes generated based on support features with query features to perform segmentation. For example, FWB [49] enhances the activation of foregrounds to improve the discrimination of features and assist in the similarity calculation between prototypes and query features. PANet [30] segments the query image with the support prototype. In addition, it segments the support set by using the initial prediction mask and the query image to align the prototype. CANet [31] introduces an idea of extending the prototype and fusing it with query features, a concept widely accepted in follow-up studies. PFENet [33] utilizes prior knowledge to mitigate model bias toward known classes. PPNet [50], PMMs [51], and ASGNet [52] associate different regions of an image into multiple prototypes to capture diverse object features and extract richer information from limited samples. DCP [53] performs mask prediction on the support image, divides the predicted mask into multiple subprototypes, and fuses similar prototypes with a parallel decoder. BAM [54] uses base learners to predict the base classes to refine the final segmentation results while predicting new classes. NTRENet [55] eliminates invalid features and confounding information via learning background and object prototypes. HSNet [56] constructs hypercorrelations via efficient 4-D convolution and emphasizes the correspondence between query and support images. HDMNet [57] combines Transformer [58] and correlation distillation to mine semantic relations. MIANet [59] integrates semantic label and instance information to segment targets. Although these methods perform well in FSS of natural images, they have limitations when applied to RSIs due to the latter's unique characteristics.

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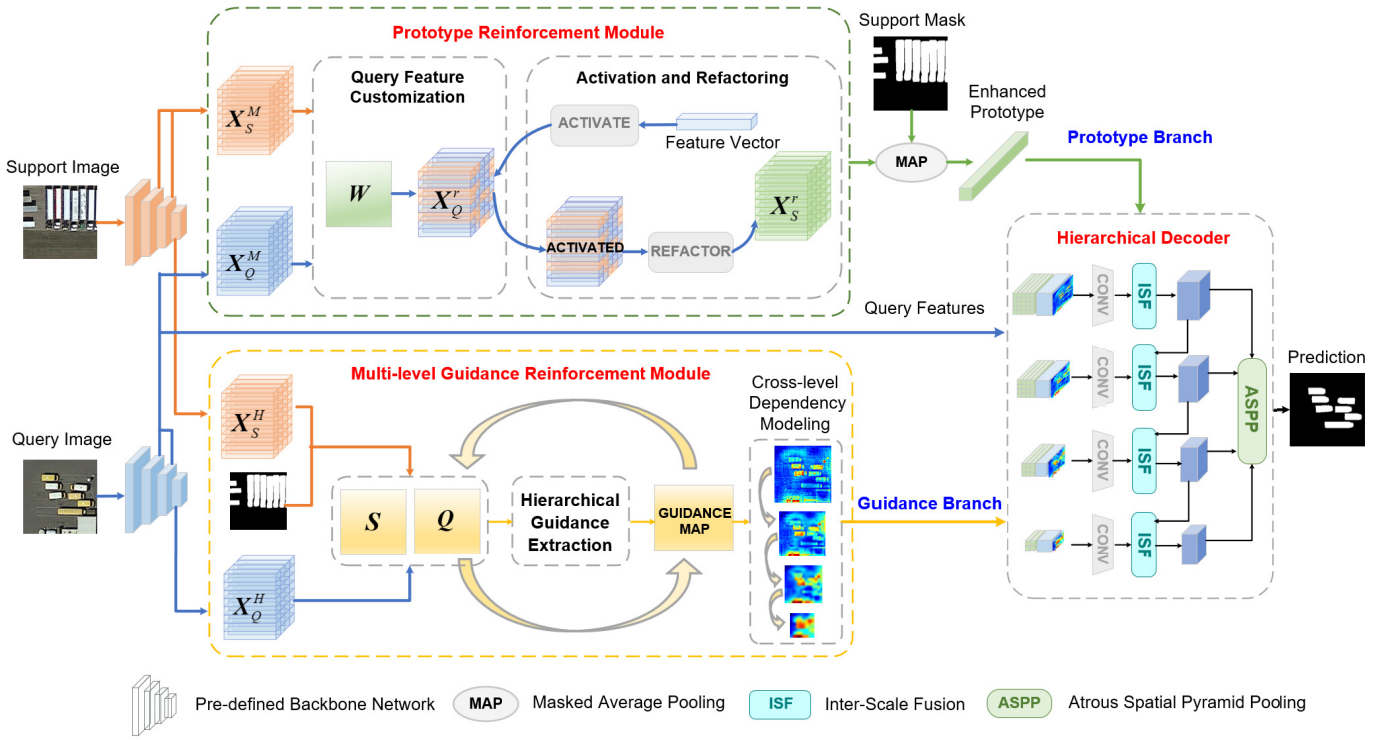


Fig. 3. Overall architecture of the proposed TBRNet. The backbone network extracts feature maps across multiple levels for the support and query images, respectively. The mid-level support and query features are fed into the PRM on the prototype branch to generate the enhanced prototypes. The high-level ones are fed into the MGRM on the guidance branch to generate multilevel guidance maps. A hierarchical decoder is adopted to seamlessly integrate the enhanced prototypes and multilevel guidance maps with the query features to yield a precise segmentation prediction of the query image.

Regarding RSIs, numerous studies on few-shot learning have been carried out in tasks such as object detection [60], [61], [62] and scene classification [63], [64], [65]. Nevertheless, research on the semantic segmentation of RSIs remains relatively scarce. For instance, SDM [35] generates prototypes by leveraging the foreground and background of support images and measures pixel-level similarity between prototypes and query features to achieve matching. PCFNet [66] evaluates correlations between support and query features at different levels, thereby enhancing semantic associations. SCCNet [67] constructs cross correlation between support and query features and self-correlation of query features to generate prediction masks. FRINet [36] acquires support information of the same category from different directions to rotate, matching the query image. DSMF-Net [37] constructs scale-aware graph prototypes via graph convolutional networks (GCNs). MGCL [68] employs oversegmented masks to ensure intramask consistency. HSE [38] enables visual support features to interact with class description embeddings spatially through self-attention.

Considering the unique characteristics of RSIs, we devise a novel yet powerful FSS framework for RSIs and verify its distinctive merits over previous methods.

III. Proposed Method

A. Problem Definition

FSS for RSIs aims to annotate potential objects in query images with only a few labeled support images. Normally, the dataset is divided into a training set $\mathcal{D}_{\text{train}}$ and a testing set $\mathcal{D}_{\text{test}}$. The categories within $\mathcal{D}_{\text{train}}$ are labeled as $\mathcal{C}_{\text{train}}$, while those in $\mathcal{D}_{\text{test}}$ are denoted as $\mathcal{C}_{\text{test}}$. It should be emphasized that $\mathcal{C}_{\text{train}}$ and $\mathcal{C}_{\text{test}}$ are mutually exclusive, i.e., $\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{train}} = \emptyset$. The model is trained on $\mathcal{D}_{\text{train}}$ to capture underlying data patterns and store learned parameters. Its generalization ability is evaluated by applying it to $\mathcal{D}_{\text{test}}$ and observing prediction results for novel categories in $\mathcal{C}_{\text{test}}$.

Conventional supervised methods struggle to generalize to novel classes in RSIs due to labeled data scarcity and domain shift. To bridge this gap, we leverage the metalearning paradigm [31], which enables rapid adaptation by simulating few-shot scenarios during training.

Specifically, $\mathcal{D}_{\text{train}}$ is split into multiple episodes. For each episode, we sample $K+1$ groups of images, i.e., $\{(S_s^i, M_s^i)\}_{i=1}^K$ and $\{(Q_q, M_q)\}$, from $\mathcal{D}_{\text{train}}$. The images in these groups all contain the same category c, which is an element of $\mathcal{C}_{\text{train}}$. K is the number of support images. The K groups of images $\{(S_s^i, M_s^i)\}_{i=1}^K$ are called the support set $\mathcal{S}$. The remaining group of images $\{(Q_q, M_q)\}$ corresponds to the query set $\mathcal{Q}$. During each episodic training, the model recognizes the category $c \in \mathcal{C}_{\text{train}}$ for the query image Q_q based on the knowledge learned from the support set $\mathcal{S}$ and calculates the loss according to M_q. The testing phase is similar to the training stage. In this phase, the model is guided by the K groups of images in the support set $\mathcal{S} \in \mathcal{D}_{\text{test}}$ to perform inference on the new category $c \in \mathcal{C}_{\text{test}}$ for the query image Q_q. However, it should be noted that the model cannot access M_q in the testing phase.

B. Method Overview

Fig. 3 shows the overall architecture of our proposed TBRNet. For a support-query image pair, our model initially

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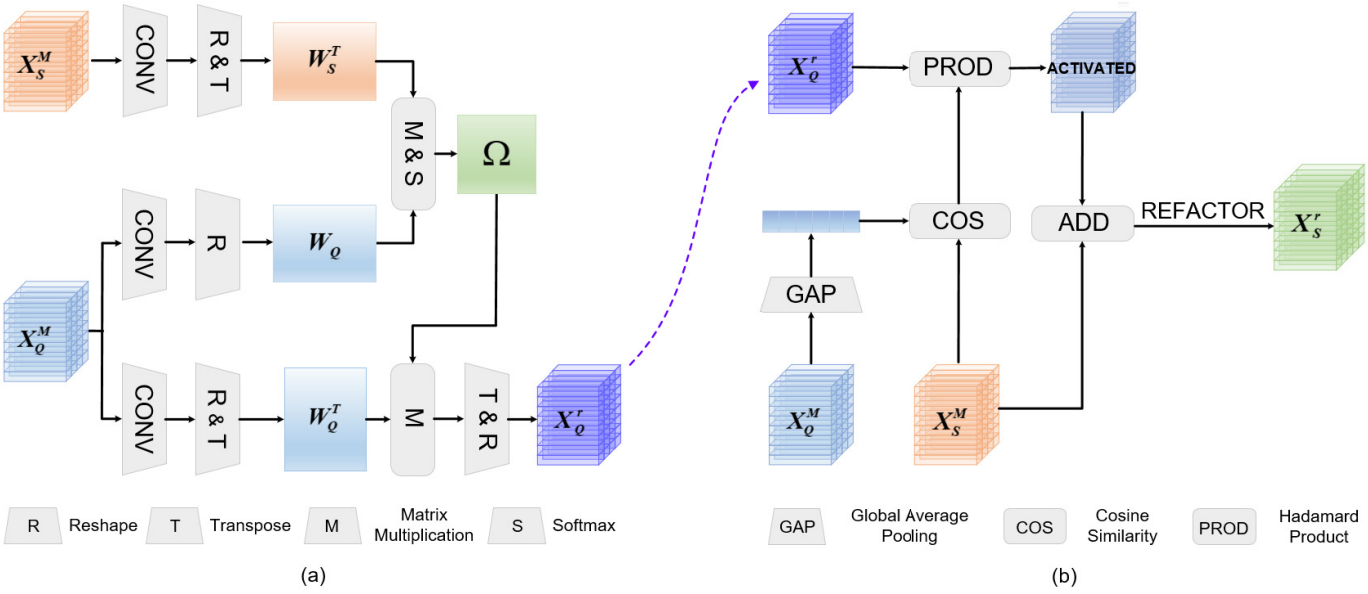


Fig. 4. Structural diagram of PRM. (a) QFC. (b) Activation and refactoring. In (a), the mid-level support and query features are used as inputs to calculate the weighted query feature X_Q^r. In (b), the weighted query feature X_Q^r is activated at first, and the activated X_Q^r is used to compute the refactored support feature X_S^r.

employs a pretrained backbone network to extract features at multiple levels for both images. To balance global context and local details, the extracted mid- and high-level features are, respectively, fed into two branches: the prototype branch and the guidance branch. These two branches are designed to generate enhanced support prototypes and multilevel guidance maps, which encode rich category-specific and contextual information. Finally, a hierarchical decoder seamlessly integrates those outputs with query features to produce a precise prediction for the query image.

Specifically, regarding the backbone network, most FSS models directly utilize a CNN pretrained on ImageNet. In contrast, to capture RSIs' complex scene contexts, our proposed TBRNet adopts a PSPNet [45]—pretrained on base classes (i.e., seen categories during training)—as the backbone. The choice is motivated by PSPNet's ability to establish long-distance dependencies, capture global image context, enabling effective generation of multilevel feature maps.

Let $X_S^M \in \mathbb{R}^{c\times h\times w}$ and $X_Q^M \in \mathbb{R}^{c\times h\times w}$ denote the extracted mid-level support and query features, respectively. To address the domain gap between support and query images, these features are fed into the prototype branch, where a PRM is designed to enhance support prototypes' sensitivity to context-relevant regions in the query image. PRM leverages the locality and specificity of mid-level features to provide customized information for refactoring support features. Finally, enhanced prototypes are obtained via MAP.

Denoted as $X_S^H \in \mathbb{R}^{c'\times h'\times w'}$ and $X_Q^H \in \mathbb{R}^{c'\times h'\times w'}$, the high-level support and query features exhibit stronger abstraction, discriminability, and robustness. To mitigate information loss in traditional single-level fusion, these are fed into the guidance branch, which houses an MGRM. MGRM uses high-level features and support masks to compute multilevel guidance maps through horizontal (same level) and vertical (cross-level) bidirectional fusion. This approach fully exploits multilevel object features, mitigates critical information loss, and strengthens predictions from a nonparametric, feature-fusion perspective.

Finally, with prototypes encoding category specificity and guidance maps providing contextual constraints, the enhanced prototypes, multilevel guidance maps, and query features are integrated in a hierarchical decoder, generating the final segmentation predictions.

C. Prototype Reinforcement Module

Due to the significant intraclass variations in RSIs, prototypes derived from limited support images often fail to capture discriminative characteristics of target classes, especially when extreme appearance differences arise. Moreover, most existing prototype generation methods overly rely on support images, neglecting the complementary information embedded in query images—information that is crucial for adapting to unseen variations.

To address this, we propose a PRM that dynamically weights the interactive features between support and query images. Specifically, PRM first generates customized query features by emphasizing context-relevant regions, which are then used to refactor support features and produce enhanced, context-adaptive prototypes. By leveraging mid-level features—rich in local structural details—as inputs, PRM further incorporates object part information for novel classes, thus ensuring robust feature representation even under severe intraclass variance.

As shown in Fig. 4, PRM consists of two main components: "QFC" and "activation and refactoring." First, the features extracted from middle layers of the backbone are concatenated in the channel dimension to obtain the inputs of PRM: the

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mid-level support and query features, $X_S^M \in \mathbb{R}^{c\times h\times w}$ and $X_Q^M \in \mathbb{R}^{c\times h\times w}$.

Then, as shown in Fig. 4(a), X_S^M and X_Q^M are separately converted to 2-D feature matrices $W_S^T \in \mathbb{R}^{hw\times c}$ and $W_Q^T \in \mathbb{R}^{hw\times c}$ as follows:

$$W_S^T = \Gamma\left(\text{conv}\left(X_S^M\right)\right) \tag{1}$$
$$W_Q^T = \Gamma\left(\text{conv}\left(X_Q^M\right)\right) \tag{2}$$

where conv denotes a 1×1 convolution operator and $\Gamma\colon \mathbb{R}^{c\times h\times w} \to \mathbb{R}^{hw\times c}$ represents a function that performs both reshape and transpose operations. Furthermore, we perform a convolution on X_Q^M and reshape it to generate $W_Q \in \mathbb{R}^{c\times hw}$

$$W_Q = \Upsilon\left(\text{conv}\left(X_Q^M\right)\right) \tag{3}$$

where $\Upsilon\colon \mathbb{R}^{c\times h\times w} \to \mathbb{R}^{c\times hw}$ denotes the reshape function.

Based on W_S^T and W_Q, the weight matrix $\Omega \in \mathbb{R}^{hw\times hw}$ used for weighting the query features can be obtained as follows:

$$\Omega = \Phi\left(W_S^T, W_Q\right) \tag{4}$$

where Φ represents the matrix multiplication and softmax operations. Each value in Ω indicates the similarity between the vectors in W_S^T and W_Q. By applying Ω to W_Q^T, we generate the weighted query feature $X_Q^r \in \mathbb{R}^{c\times h\times w}$

$$X_Q^r = \text{H}\left(\Omega \otimes W_Q^T\right) \tag{5}$$

where $\otimes$ represents the matrix multiplication operation and $\text{H}\colon \mathbb{R}^{hw\times c} \to \mathbb{R}^{c\times h\times w}$ represents transpose and reshape operations. X_Q^r emphasizes similarities between support and query features, providing customized information for enhanced prototype generation.

As shown in Fig. 4(a), "QFC" generates X_Q^r, which is fed into the "activation and refactoring" component [see Fig. 4(b)]. First, to suppress background noise and highlight target regions, X_Q^M passes through a global average pooling (GAP) layer. The cosine similarity $A \in \mathbb{R}^{1\times h\times w}$ between the resulting vector and the support feature X_S^M is then calculated to activate valid regions in X_Q^r. Finally, the activated X_Q^r is summed with X_S^M to obtain the refactored feature $X_S^r \in \mathbb{R}^{c\times h\times w}$. The process can be written as

$$A = F_{\text{cosine}}\left(\text{avgpool}\left(X_Q^M\right), X_S^M\right) \tag{6}$$
$$X_S^r = X_S^M + A \odot X_Q^r \tag{7}$$

where F_{cosine} indicates the cosine similarity function and $\odot$ denotes the Hadamard product.

After obtaining X_S^r, the enhanced prototype $pro \in \mathbb{R}^{c\times 1\times 1}$ is computed via MAP

$$pro = \text{MAP}\left(X_S^r, M_S\right) \tag{8}$$

where $M_S \in \mathbb{R}^{1\times h\times w}$ is the support mask.

The enhanced prototype from PRM integrates target-category-specific information from query features, reducing background noise and enhancing the discriminative power of prototype representation.

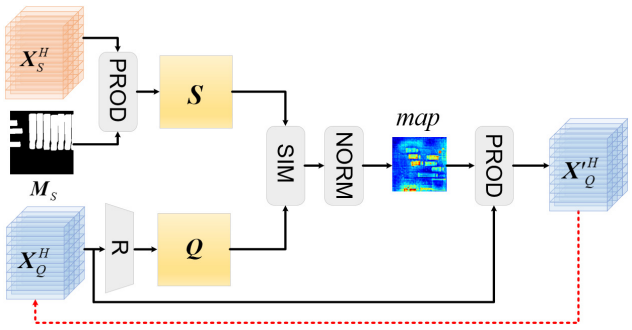


Fig. 5. Generation process of a guidance map for a specific level, where the guidance map is denoted as *map*.

D. Multilevel Guidance Reinforcement Module

The coexistence of multiple target categories in RSIs can lead to generalization failure. To address this challenge, we propose an MGRM.

Unlike PRM's focus on mid-level local structures, MGRM utilizes high-level features for stronger semantic specificity. Specifically, we first extract semantic correlations between the high-level features of the query and support images to generate guidance information for the query image—serving as a rough activation map of target objects. To capture richer prior information for objects of different scales and shapes, MGRM incorporates a multilevel strategy that generates guidance maps at multiple levels through horizontal (same level) and vertical (cross-level) bidirectional fusion. This multilevel strategy models cross-level semantic dependencies and preserves more distinctive details, aiding precise identification of object classes in the query image.

Fig. 5 illustrates the generation process of a guidance map for a specific level, taking three inputs: the high-level query feature $X_Q^H \in \mathbb{R}^{c'\times h'\times w'}$, the high-level support feature $X_S^H \in \mathbb{R}^{c'\times h'\times w'}$, and the corresponding support mask $M_S \in \mathbb{R}^{1\times h'\times w'}$.

First, we apply the Hadamard product to X_S^H and M_S to eliminate background interference in support features

$$S = \Gamma\left(X_S^H \odot M_S\right) \tag{9}$$

where $S \in \mathbb{R}^{h'w'\times c'}$, $\odot$ denotes the Hadamard product, and Γ denotes reshape and transpose operations. X_Q^H is also reshaped into $Q \in \mathbb{R}^{c'\times h'w'}$.

In the horizontal direction (same-level), we calculate the similarity between S and Q

$$sim = \frac{SQ}{\|S\|\,\|Q\|} \in \mathbb{R}^{h'w'\times h'w'}. \tag{10}$$

Subsequently, we generate the guidance map ***map*** by reshaping $sim' = \max(sim) \in \mathbb{R}^{1\times h'w'}$ into $map \in \mathbb{R}^{1\times h'\times w'}$. The final normalized guidance map is obtained by a min–max scheme

$$map = \frac{map - \min(map)}{\max(map) - \min(map) + \varsigma} \in \mathbb{R}^{1\times h'\times w'} \tag{11}$$

where the value $1e^{-7}$ is set for ς to prevent division by zero.

To integrate multiscale cues for RSI objects such as small vehicles and large fields, we first refine the current guidance

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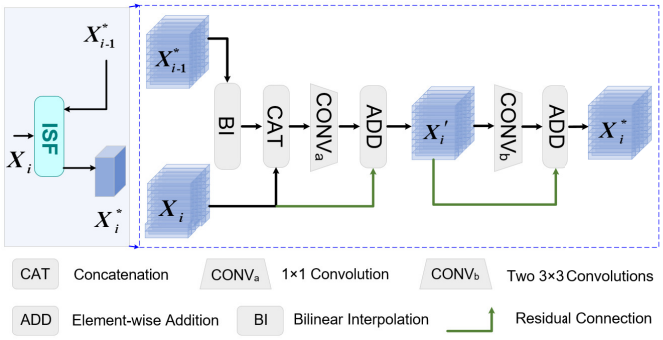


Fig. 6. Architecture of the ISF block for cross-scale feature integration with residual learning.

map ***map*** by multiplying it with X_Q^H for vertical (cross-level) guidance

$$X'^H_Q = map \odot X_Q^H \in \mathbb{R}^{c' \times h' \times w'}. \tag{12}$$

We then downscale X'^H_Q, X_S^H, and M_S to the next level and repeat the guidance generation process. After processing four levels, we obtain a set of guidance maps $\{map_i\}_{i=1}^4$, generated in a top-down manner with resolutions decreasing hierarchically from high to low. This design leverages multilevel guidance maps to capture complementary structural and locational information, essential for robust object representation.

Each level's guidance map reflects distinct structural and positional cues of target categories in the query image. By fusing multilevel guidance maps, enhanced prototypes, and query features together, MGRM produces stronger semantic representations for accurate segmentation.

E. Hierarchical Decoder

Given that MGRM generates guidance maps at multiple levels, we construct a hierarchical decoder (see Fig. 3) to leverage this multilevel information. In this decoder, enhanced prototypes, multilevel guidance maps, and query features are seamlessly fused to generate refined features for semantic segmentation. In addition, mid-level query features—rich in local structural details and spatially compact—are used for integration instead of high-level features, addressing spatial inconsistency between query and support images.

Specifically, at a certain level, $map \in \mathbb{R}^{1 \times h' \times w'}$ denotes the guidance map. The enhanced support prototype $pro \in \mathbb{R}^{c \times 1 \times 1}$ is first expanded to $pro \in \mathbb{R}^{c \times h' \times w'}$, and the mid-level query feature $X_Q^M \in \mathbb{R}^{c \times h \times w}$ is processed via adaptive average pooling to generate $X_Q^M \in \mathbb{R}^{c \times h' \times w'}$.

Then, concatenation and convolution operations are used to integrate $map \in \mathbb{R}^{1 \times h' \times w'}, pro \in \mathbb{R}^{c \times h' \times w'}$, and $X_Q^M \in \mathbb{R}^{c \times h' \times w'}$ along the channel dimension

$$X_i = \text{conv}\big(\text{cat}\big(pro, X_Q^M, map_i\big)\big), \quad i = 1, 2, 3, 4 \tag{13}$$

where cat denotes concatenation and *conv* is a 1×1 convolution yielding the merged feature with 256 output channels.

Subsequently, we construct an interscale fusion block (ISF; see Fig. 6) to refine the preliminary fusion feature X_i.

Let X_i denote the feature from the current level and X_{i-1}^* be the refined feature from the previous level. To ensure spatial consistency, X_{i-1}^* is first aligned to the spatial dimension of X_i via bilinear interpolation. A 1×1 convolution (denoted as conv_a) is applied to the concatenation of X_{i-1}^* and X_i, extracting cross-scale information. The output is elementwise added to X_i through a residual connection, forming the intermediate feature X_i'

$$X_i' = \text{conv}_a\big(\text{cat}\big(X_{i-1}^*, X_i\big)\big) + X_i. \tag{14}$$

Two consecutive 3×3 convolutions (collectively denoted as conv_b) process X_i' to facilitate feature interaction. A second residual connection adds the output of conv_b back to X_i', generating the refined feature X_i^*

$$X_i^* = \text{conv}_b\big(X_i'\big) + X_i'. \tag{15}$$

For simplicity, the entire ISF process is denoted as

$$X_i^* = \text{ISF}\big(X_{i-1}^*, X_i\big), \quad i = 2, 3, 4. \tag{16}$$

When auxiliary features are unavailable (e.g., the first merged feature in the top-down path), the module skips the concatenation with X_{i-1}^*, degenerating to

$$X_i^* = \text{conv}_b(X_i) + X_i, \quad i = 1. \tag{17}$$

As visualized by the looped arrows in Fig. 6, the residual connections in (14) and (15) preserve feature integrity, while the hierarchical design of ISF enables effective cross-scale information integration.

After obtaining multilevel ISF outputs $(X_1^*, X_2^*, X_3^*, X_4^*)$, we employ interpolation, concatenation, and convolutional operations to generate the refined features for semantic segmentation

$$X_Q^* = \text{conv}\big(\text{cat}\big(X_1^*, X_2^*, X_3^*, X_4^*\big)\big). \tag{18}$$

Finally, the ASPP module processes X_Q^* to produce the final prediction.

F. Loss Function

Building on the hierarchical decoder's multilevel feature fusion, we design a loss function to address two critical challenges in RSI segmentation: 1) category imbalance and 2) multiscale object representation.

To tackle these challenges, we employ a pixelwise cross-entropy loss to measure the discrepancy between the predicted mask and the ground truth (GT).

The total loss L_{total} of TBRNet consists of two components: L_{main} and L_{decoder}. L_{main} quantifies the difference between the final prediction Y and the GT GT, while L_{decoder} accounts for the loss from intermediate supervision in the decoding processing. During this process, multiple hierarchical predictions are generated, and their discrepancies with the GT are denoted as $\{L_{\text{PG}}^k\}_{k=1}^n$

$$L_{\text{main}} = \text{CrossEntropyLoss}(Y, GT) \tag{19}$$

$$L_{\text{decoder}} = \frac{\alpha}{n} \sum_{k=1}^{n} L_{\text{PG}}^k \tag{20}$$

$$L_{\text{total}} = L_{\text{main}} + L_{\text{decoder}} \tag{21}$$

where $n = 4$ (the number of intermediate predictions) and $\alpha = 1$ are set empirically.

The execution flow for TBRNet is presented in Algorithm 1.

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Algorithm 1 Training and Testing TBRNet

Input:

S, one support image and its mask M_S for class c in each episode τ

Q, the query image in each episode τ

Output:

M_Q, a segmentation mask for the query image Q θ, network parameters

Extract mid-level and high-level support and query features $X_S^M, X_S^H, X_Q^M, X_Q^H$ from S and Q via pre-trained model;

*** **Prototype Reinforcement Module** ***

Calculate weight matrix Ω using (1)–(4);

Generate weighted query features X_Q^r by (5);

Activate valid regions in X_Q^r and refactor support features X_S^r following (6)–(7);

Construct enhanced prototype *pro* using masked average pooling by (8);

*** **Multi-Level Guidance Reinforcement Module** ***

$levels = \{l_1, l_2, l_3, l_4\}$

for each level i in *levels* **do**

  Calculate the guidance map map_i on this level as in (9)–(11);

  Multiply map_i with X_Q^H to obtain query feature X'^H_Q for the next level following (12);

end for

*** **Hierarchical Decoder** ***

for each level i in *levels***do**

  Generate fusion feature X_i by integrating map_i, *pro* and X_Q^M as in (13);

  Compute refined feature X_i^* by fusing X_i and X_{i-1}^* from preceding level as in (14)–(17);

end for

Generate final features X_Q^* for segmentation by (18);

Predict a mask M_Q for the query image Q;

*** **Meta Training/Testing** ***

if Training_mode **then**

  Compute the loss L_{total} as in (19)–(21);

  Update θ to minimize the loss L_{total};

end if

IV. Experiments

A. Experimental Datasets and Metrics

Currently, there are extremely limited datasets specially designed for FSS in RSIs. To ensure a fair comparison, we meticulously select the iSAID-5^i dataset [34] and the DLRSD-5^i dataset [69] as benchmarks for our experiments.

iSAID-5^i and DLRSD-5^i, widely adopted in prior few-shot RS studies, are adapted from the general datasets iSAID and DLRSD—two of the few publicly available resources capable of supporting such tasks. Given the absence of standardized few-shot benchmarks, the research community has established protocols (e.g., n-way k-shot splits) to adapt these general datasets to few-shot scenarios. This approach ensures consistency with methodological norms in the field while leveraging the rich data characteristics of iSAID and DLRSD.

Through this adaptation process, iSAID-5^i and DLRSD-5^i have emerged as de facto benchmarks for FSS in RS, with numerous prior studies validating their methods on these datasets. By following the established dataset protocols, our results maintain direct comparability with the state of the art, facilitating rigorous cross-study evaluation and ensuring the reproducibility of our findings.

While both datasets adhere to common few-shot protocols, they exhibit distinct characteristics that test different aspects of model performance. The iSAID-5^i dataset contains a large number of high-resolution RSIs covering diverse land cover categories and complex scenes, making it well-suited for few-shot learning tasks. The DLRSD-5^i dataset, on the other hand, presents different challenges, such as diverse category compositions and variable scene characteristics, further verifying the generalization capability of our model.

The specific information of these two datasets is introduced as follows.

The iSAID-5^i dataset, which consists of 2806 high-resolution RSIs, contains 655 451 object instances of 15 categories. For cross-validation purposes, we divide these 15 categories into threefold: in each fold, ten categories are used for training, while the remaining five categories are employed for validation, as indicated in Table I. Following most existing methods, we adopt the same cropping strategy (cropping images into 256×256 patches). Consequently, a total of 18076 images are collected for training, and 6363 samples are used for testing.

The DLRSD-5^i dataset is derived from the original DLRSD dataset, a publicly available benchmark for evaluating multi-label RS tasks [70]. DLRSD consists of 2100 images across 17 object categories, each fixed at 256×256. For our FSS task, we select 15 classes from DLRSD to construct DLRSD-5^i, which is divided into threefold (see Table II). To prevent information leakage, images containing testing categories are filtered out during the training phase.

Following [36] and [37], we select mean intersection over union (mIoU) as the primary evaluation index. mIoU measures per-class segmentation quality, offering a more comprehensive and objective metric than foreground–background IoU (FB-IoU) [71]—especially critical for class-imbalance scenarios. mIoU is defined as $(1/n)\sum_{i=1}^{n} \mathrm{IoU}_i$, where n represents the number of classes and IoU_i denotes the IoU value for class i.

B. Implement Details

All experiments are implemented using PyTorch on a server equipped with a server equipped with a NVIDIA RTX 3090 GPU.

In our TBRNet, we construct PSPNet with VGG-16 or ResNet-50 as the backbone network for deep feature extraction. In the backbone network, we initialize the VGG-16 or ResNet-50 part with publicly available ImageNet pretrained weights, while the other part is initialized with random weights. The initialized backbone network is then fine-tuned on the iSAID-5^i and DLRSD-5^i datasets, respectively, to adapt to RS domain characteristics. This enables the backbone to fully acquire base-class knowledge and extract unique features of RSIs. After the backbone network is fine-tuned, during the

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TABLE I
CROSS-VALIDATION DIVISION SETTINGS FOR iSAID-5^i

Fold	Base Classes (Train)	Target Classes (Test)
Fold-0	ground track field (C6), bridge (C7), large vehicle (C8), small vehicle (C9), helicopter (C10), swimming pool (C11), roundabout (C12), soccer ball field (C13), plane (C14), harbor (C15)	ship (C1), storage tank (C2), baseball diamond (C3), tennis court (C4), basketball court (C5)
Fold-1	ship (C1), storage tank (C2), baseball diamond (C3), tennis court (C4), basketball court (C5), swimming pool (C11), roundabout (C12), soccer ball field (C13), plane (C14), harbor (C15)	ground track field (C6), bridge (C7), large vehicle (C8), small vehicle (C9), helicopter (C10)
Fold-2	ship (C1), storage tank (C2), baseball diamond (C3), tennis court (C4), basketball court (C5), ground track field (C6), bridge (C7), large vehicle (C8), small vehicle (C9), helicopter (C10)	swimming pool (C11), roundabout (C12), soccer ball field (C13), plane (C14), harbor (C15)

TABLE II
CROSS-VALIDATION DIVISION SETTINGS FOR DLRSD-5^i

Fold	Base Classes (Train)	Target Classes (Test)
Fold-0	court (C6), dock (C7), grass (C8), mobile home (C9), pavement (C10), sand (C11), ship (C12), tanks (C13), trees (C14), water (C15)	airplane (C1), bare soil (C2), buildings (C3), cars (C4), chaparral (C5)
Fold-1	airplane (C1), bare soil (C2), buildings (C3), cars (C4), chaparral (C5), sand (C11), ship (C12), tanks (C13), trees (C14), water (C15)	court (C6), dock (C7), grass (C8), mobile home (C9), pavement (C10)
Fold-2	airplane (C1), bare soil (C2), buildings (C3), cars (C4), chaparral (C5), court (C6), dock (C7), grass (C8), mobile home (C9), pavement (C10)	sand (C11), ship (C12), tanks (C13), trees (C14), water (C15)

training phase of TBRNet, the pretrained backbone weights are frozen to accelerate model training and keep the model focused on the task at hand.

We adopt SGD as the optimizer, where the momentum is set to 0.9, the weight decay is set to $1e^{-4}$, and the "poly" learning rate policy is utilized to gradually decrease the learning rate [29], [33], [38]. For iSAID-5^i, the initial learning rate is set to 0.005; for DLRSD-5^i, the initial learning rate is set to 0.007. Meanwhile, we use different data augmentation techniques such as random scaling, random rotation, Gaussian blurring, horizontal flipping, and cropping to increase the diversity of the training data and improve the generalization ability of the model. TBRNet is trained in an episodic manner for 100 epochs with a batch size of 6; at the same time, model testing is carried out by randomly sampling 1000 episodes from the testing set.

C. Ablation Studies

To evaluate the effectiveness of various modules in our TBRNet, a series of ablation studies is conducted. The experiments are performed on the iSAID-5^i dataset, with ResNet-50-based PSPNet as the backbone for the one-shot task.

1) Ablation Studies of Different Modules in TBRNet: First, we discuss the role of MGRM. Since MGRM mines guidance knowledge to identify target categories, it effectively alleviates the generalization failure issue caused by multicategory coexistence.

As displayed in the first two rows of Table III, the MGRM-integrated model (second row) outperforms the baseline (first row) across Fold-0, Fold-1, and Fold-2. The averaged class mIoU rises to 42.47% (a 2.39% improvement over the baseline).

TABLE III
ABLATION STUDIES OF MGRM AND PRM USING MIOU (%) EVALUATION METRIC

MGRM	PRM	Fold-0	Fold-1	Fold-2	Mean
		46.59	33.60	40.05	40.08
✓		48.49	37.86	41.05	42.47**(+2.39)**
	✓	47.80	38.10	41.37	42.42**(+2.34)**
✓	✓	50.65	38.20	43.08	43.98**(+3.90)**

Fig. 7 visualizes MGRM-generated multilevel guidance maps, demonstrating its ability to highlight spatial localizations of diverse categories. By calculating precise multilevel guidance, MGRM reduces the risk of generalization collapse.

Next, we explore PRM's effectiveness (third row of Table III). The PRM-integrated model significantly outperforms the baseline, with mIoU increases of 1.21% (Fold-0), 4.5% (Fold-1), and 1.32% (Fold-2). These results confirm that PRM effectively integrates query features with prototypes, strengthening prototype representation and improving segmentation accuracy in complex scenarios.

Finally, combining MGRM and PRM in TBRNet (last row of Table III) further boosts performance, achieving 3.9%, 1.51%, and 1.56% improvements over the baseline, MGRM-only, and PRM-only models, respectively. This mutual

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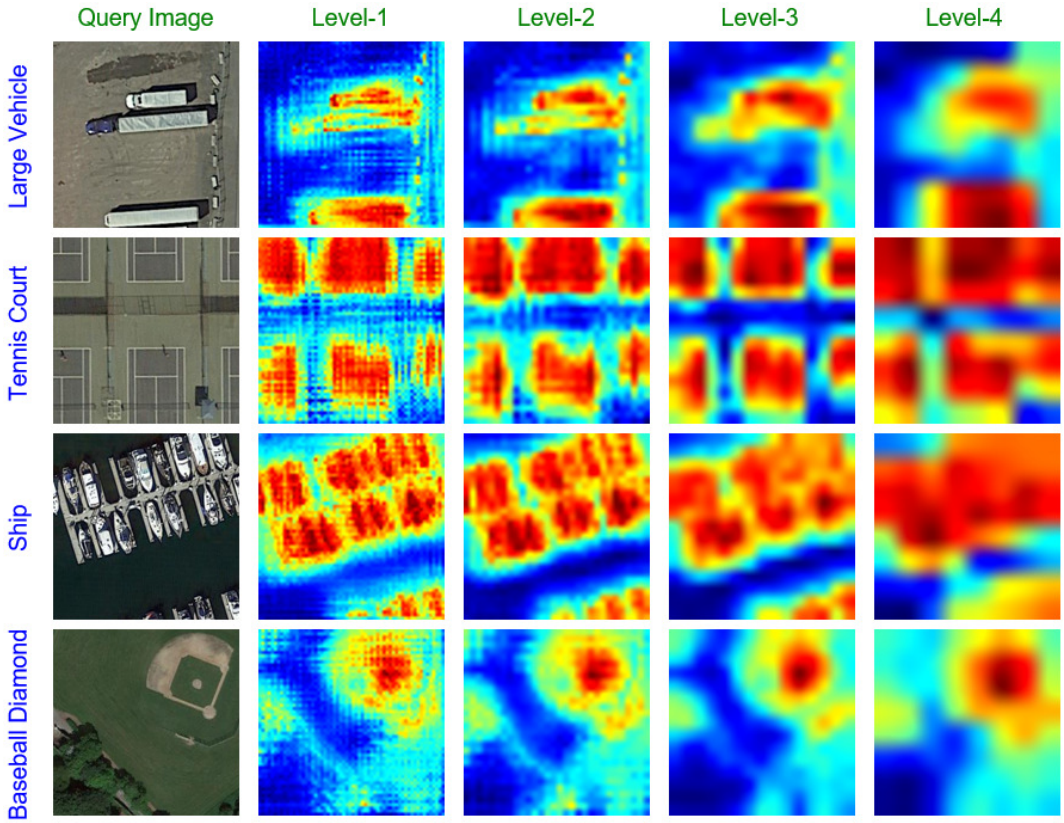


Fig. 7. Illustration of the effect of MGRM.

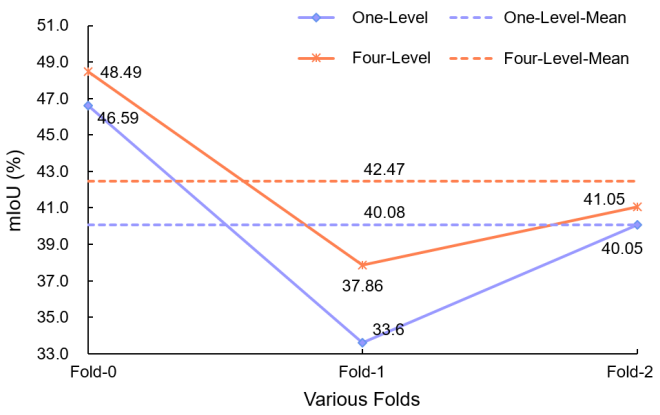


Fig. 8. Comparison between multilevel MGRM and single-level MGRM.

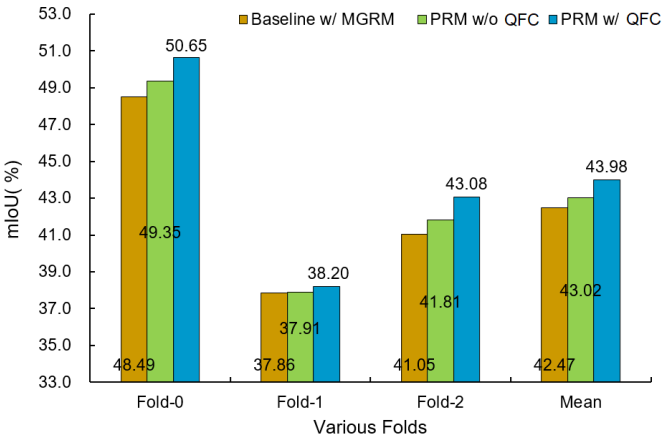


Fig. 9. Ablation studies of QFC.

complementarity—MGRM providing rich prior knowledge and PRM enhancing prototype representation—drives the network's significant performance gain.

2) Ablation Studies of Key Components in MGRM: MGRM's multilevel guidance map generation method is compared with a single-level variant (as shown in Fig. 8).

As can be seen from Fig. 8, the multilevel strategy yields a 2.39% mIoU improvement, with a 4.26% gain on Fold-1, confirming that interlevel communication reduces feature loss and enhances global information capture.

3) Ablation Studies of Key Components in PRM: QFC is crucial to PRM. Fig. 9 compares three variants: "baseline + MGRM" (baseline w/MGRM), "baseline + MGRM + PRM without QFC" (PRM w/o QFC), and "baseline + MGRM + PRM with QFC" (PRM w/QFC).

Without QFC, mIoU scores are 49.35% (Fold-0), 37.91% (Fold-1), and 41.81% (Fold-2). With QFC, they rise to 50.65%, 38.20%, and 43.08%, respectively. Furthermore, QFC enables PRM to leverage query-image information, enhancing prototype expressiveness and segmentation performance compared to support-feature-only approaches.

D. Quantitative Comparisons With State-of-the-Art

To demonstrate TBRNet's outstanding performance in FSS for RSIs, we compare it with state-of-the-art FSS models,

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TABLE IV

COMPARISON OF ONE- AND FIVE-SHOT PERFORMANCES OF DIFFERENT MODELS ON iSAID-5^i. THE BOLD RED AND GREEN FONTS INDICATE THE TOP TWO PERFORMANCES, RESPECTIVELY

Method	Backbone	1-shot				5-shot			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
PMMs [51]	VGG-16	20.87	16.07	24.65	20.53	23.31	16.61	27.43	22.45
HSNet [56]		23.41	21.61	22.01	22.34	28.76	27.15	25.90	27.27
PFENet [33]		16.68	15.30	27.87	19.95	18.46	18.39	28.81	21.89
DCP [53]		23.52	18.98	28.84	23.78	25.45	20.70	29.41	25.19
NTRENet [55]		9.15	13.57	9.81	10.84	6.95	20.06	11.75	12.92
SDM [35]		29.24	20.80	34.73	28.26	36.33	27.98	42.39	35.57
HDMNet [57]		19.11	14.23	22.98	18.77	19.29	14.66	23.38	19.11
SCCNet [67]		35.71	30.33	36.68	34.24	40.40	32.56	39.31	37.42
MGCL [68]		38.24	28.61	40.21	35.69	**43.81**	31.07	**48.31**	41.06
FRINet [36]		**42.86**	**36.73**	**42.51**	**40.70**	**43.80**	**37.18**	**45.41**	**42.13**
DSMF-Net [37]		31.25	24.35	31.04	28.88	40.73	26.92	38.05	35.23
HSE [38]		37.18	29.79	36.04	34.34	40.87	30.20	43.44	38.17
TBRNet (Ours)		**42.72**	**35.39**	**41.21**	**39.77**	43.32	**39.31**	45.24	**42.62**
PMMs [51]	ResNet-50	19.02	18.51	28.42	21.98	20.89	20.87	31.23	24.33
HSNet [56]		30.82	22.91	32.94	28.89	32.94	29.73	40.76	34.48
PFENet [33]		18.75	17.24	22.09	19.36	19.57	18.43	26.14	21.38
DCP [53]		26.66	24.56	21.68	23.34	28.81	24.21	31.73	28.25
NTRENet [55]		8.15	18.59	11.85	12.86	9.24	20.60	10.78	13.54
SDM [35]		34.29	22.25	35.62	30.72	39.88	30.59	45.70	38.72
HDMNet [57]		36.04	32.12	34.87	34.34	39.95	36.01	40.68	38.88
SCCNet [67]		40.74	31.25	**46.40**	39.45	44.27	31.62	50.45	42.11
MGCL [68]		42.77	30.59	**46.39**	39.92	**49.14**	32.58	**52.75**	**44.82**
FRINet [36]		**46.50**	**36.94**	43.93	**42.59**	48.85	**38.05**	46.46	44.45
DSMF-Net [37]		39.03	25.92	40.57	35.17	43.69	30.81	42.32	38.94
HSE [38]		44.03	35.87	39.26	39.72	47.03	36.75	45.66	43.15
TBRNet (Ours)		**50.65**	**38.20**	43.08	**43.98**	**54.85**	**39.81**	**51.25**	**48.64**

including PMMs [51], HSNet [56], PFENet [33], DCP [53], NTRENet [55], SDM [35], HDMNet [57], SCCNet [67], FRINet [36], DSMF-Net [37], HSE [38], MGCL [68], and PCFNet [66].

Experimental results on the iSAID-5^i dataset are shown in Table IV, where the best performance in each comparison is bolded red, and the suboptimal performances are bolded green. Evidently, TBRNet outperforms most methods.

1) With the VGG-16 backbone, TBRNet achieves a mean mIoU of 39.77% on the one-shot task, slightly lower than FRINet's 40.70%.
2) In the five-shot setting, FRINet attains 42.13% mean mIoU, while TBRNet achieves the optimal 42.62% (a 0.49% improvement). MGCL obtains the best results on Fold-0 (43.81%) and Fold-2 (48.31%).
3) Using the ResNet-50 backbone, TBRNet achieves the highest mean mIoU (43.98% in one-shot and 48.64% in five-shot).
4) TBRNet also leads in twofold in the five-shot evaluation, 54.85% on Fold-0 and 39.81% on Fold-1, while MGCL achieves 52.75% on Fold-2.

Table V details per-class performance on iSAID-5^i, where TBRNet excels in most cases.

1) With VGG-16, it achieves the best mIoU of 52.95% for the soccer ball field (C13) class.
2) With ResNet-50, it tops the tennis court (C4) class with 59.73% mIoU.
3) Exceptional performance is also observed for the baseball diamond (C3) and swimming pool (C11).
4) For small objects like a storage tank (C2), TBRNet's 47.07% mIoU represents a 6.86% improvement over HSE, demonstrating its ability to mine tiny object features.
5) For visually similar classes (large vehicle C8 and small vehicle C9), TBRNet shows a strong discriminative power, outperforming methods that struggle with appearance ambiguity.
6) In complex environments, TBRNet improves segmentation for classes like bridge (C7), achieving 47.34%, mIoU, and showcasing anti-interference capability.
7) Besides TBRNet, the top suboptimal performers are FRINet and HSE.

Table VI presents results on DLRSD-5^i, where TBRNet handles category variations effectively. With VGG-16/ResNet-50 backbones, it achieves 10.31%/5.98% mean mIoU improvements in one-shot tasks and 8.29%/5.11% gains in five-shot tasks compared to suboptimal methods (PCFNet and MGCL).

E. Qualitative Comparison With State-of-the-Art

In addition to the above quantitative comparisons, we also conduct a more intuitive comparison between our network and other state-of-the-art methods. Fig. 10 presents the qualitative results on the iSAID-5^i dataset, where segmented objects are labeled in red, and notable improvements are highlighted with colored boxes.

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TABLE V

COMPARISON OF CLASS-BY-CLASS PERFORMANCE OF DIFFERENT MODELS FOR THE ONE-SHOT TASK ON iSAID-5^i. THE BOLD RED AND GREEN FONTS INDICATE THE TOP TWO PERFORMANCES, RESPECTIVELY

Method	Backbone	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
PMMs [51]	VGG-16	12.84	23.06	27.81	26.65	13.97	18.48	13.76	25.58	11.68	10.88	28.52	34.62	34.15	16.32	9.64
PFENet [33]		3.94	4.70	30.95	29.21	14.59	15.01	14.63	27.68	10.13	9.06	29.40	**47.89**	25.23	25.20	11.63
DCP [53]		19.70	2.71	34.21	38.71	22.28	19.84	3.77	35.78	18.26	17.26	20.08	28.05	46.50	29.97	19.61
NTRENet [55]		0.30	36.90	5.12	2.14	1.59	2.21	24.23	14.70	13.45	13.29	9.31	7.49	14.48	4.89	12.85
SDM [35]		26.55	30.57	33.01	35.06	21.03	20.54	28.89	32.45	9.62	12.59	30.79	43.52	49.47	27.51	**22.37**
HDMNet [57]		17.66	0.38	28.60	27.37	21.55	26.64	12.74	15.21	5.43	11.15	12.08	30.57	43.88	11.92	16.43
FRINet [36]		**37.26**	18.21	**43.32**	**60.56**	**54.96**	26.86	**39.84**	**52.26**	**31.29**	**33.39**	28.57	**65.20**	**51.08**	**48.27**	19.41
HSE [38]		**37.56**	**40.21**	40.65	36.20	31.28	**34.30**	38.24	33.26	20.01	**23.14**	**32.70**	43.45	47.63	30.89	**25.53**
TBRNet		34.37	**47.07**	**48.35**	**49.35**	**34.48**	**29.72**	**47.34**	**52.94**	**25.00**	21.93	**51.36**	44.50	**52.95**	**41.98**	15.24
PMMs [51]	ResNet-50	12.93	17.92	41.97	7.43	14.87	13.24	32.67	26.08	4.17	16.37	12.26	**61.49**	38.37	17.92	12.02
PFENet [33]		2.98	6.30	33.29	34.49	16.71	8.68	8.67	34.62	18.19	16.08	28.58	24.47	7.37	39.76	10.31
DCP [53]		21.14	0.50	35.15	46.52	30.51	27.75	19.87	39.44	13.37	22.37	28.55	14.79	24.97	33.20	7.77
NTRENet [55]		0.50	27.26	8.59	2.72	2.17	9.68	33.36	24.28	13.86	11.80	6.63	8.29	14.37	6.92	23.03
SDM [35]		37.66	34.37	34.45	39.81	25.14	16.77	34.53	30.50	12.42	17.02	20.69	56.83	42.80	40.52	17.26
HDMNet [57]		40.67	3.58	40.74	**56.69**	38.53	24.16	40.05	53.05	24.42	18.92	**39.03**	42.44	28.73	**51.24**	12.92
FRINet [36]		35.80	**52.96**	**43.75**	53.36	**46.61**	24.99	**48.87**	**63.38**	27.58	19.89	34.30	53.44	**55.52**	**52.44**	**23.94**
HSE [38]		**45.67**	46.21	43.34	41.02	43.91	**33.40**	40.08	36.13	**39.30**	**30.44**	28.58	50.61	45.89	43.50	**27.72**
TBRNet		**41.65**	**51.83**	**51.27**	**59.73**	**48.77**	**37.09**	**42.21**	**56.82**	**30.86**	**24.03**	**49.56**	**57.00**	**50.63**	44.03	14.16

TABLE VI

COMPARISON OF ONE- AND FIVE-SHOT PERFORMANCES OF DIFFERENT MODELS ON DLRSD-5^i. THE BOLD RED AND GREEN FONTS INDICATE THE TOP TWO PERFORMANCES, RESPECTIVELY

Method	Backbone	1-shot				5-shot			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
HSNet [56]	VGG-16	20.26	25.80	16.95	21.00	26.02	29.92	27.07	27.67
PFENet [33]		17.08	15.85	13.70	15.54	21.25	17.40	18.10	18.92
PCFNet [66]		**30.75**	**36.38**	**40.76**	**35.96**	**35.64**	**41.57**	**51.18**	**42.80**
TBRNet (Ours)		**45.04**	**43.83**	**49.95**	**46.27**	**47.72**	**48.30**	**57.27**	**51.09**
HSNet [56]	ResNet-50	24.61	29.45	28.24	27.43	29.67	31.95	36.93	32.85
PFENet [33]		21.98	20.67	11.17	17.94	24.12	24.76	16.01	21.63
PCFNet [66]		**31.09**	37.52	41.89	36.83	**36.67**	42.70	**54.20**	44.52
MGCL [68]		30.37	**50.57**	**43.01**	**41.32**	33.64	**54.84**	52.34	**46.94**
TBRNet (Ours)		**46.37**	**44.72**	**50.81**	**47.30**	**48.59**	**49.46**	**58.11**	**52.05**

Some interesting observations can be obtained as follows.

1) *Subtle Class Discrimination (First Column):* State-of-the-art methods such as SCCNet, FRINet, and MGCL misclassify basketball courts and baseball diamonds as tennis courts. In contrast, TBRNet precisely distinguishes the subtle structural differences among these classes, successfully separating easily confused categories.
2) *Multicategory Coexistence (Second Column):* SCCNet, FRINet, and MGCL incorrectly label small vehicles as a part of the bridge. TBRNet, however, effectively differentiates small vehicles from the bridge, demonstrating its capability to handle complex multicategory scenes.
3) *Complete Object Segmentation (Third and Sixth Columns):* Compared with baselines, TBRNet segments more holistic object regions, a critical advantage for FSS in RSI, where partial objects are common.
4) *Small Object Robustness (Fourth Column):* SCCNet and FRINet miss small vehicle regions and produce false detections, while MGCL omits some object areas. TBRNet accurately segments small vehicles without false positives, showcasing superior precision for tiny objects.
5) *Shape Variation Handling (Fifth Column):* Support and query images of swimming pools exhibit shape differences, leading competing models to fail to produce satisfactory results. TBRNet, however, accurately identifies the target with only one support sample, highlighting its adaptability to intraclass variations.

In summary, TBRNet's remarkable performance stems from key design choices. First, MGRM captures multilevel

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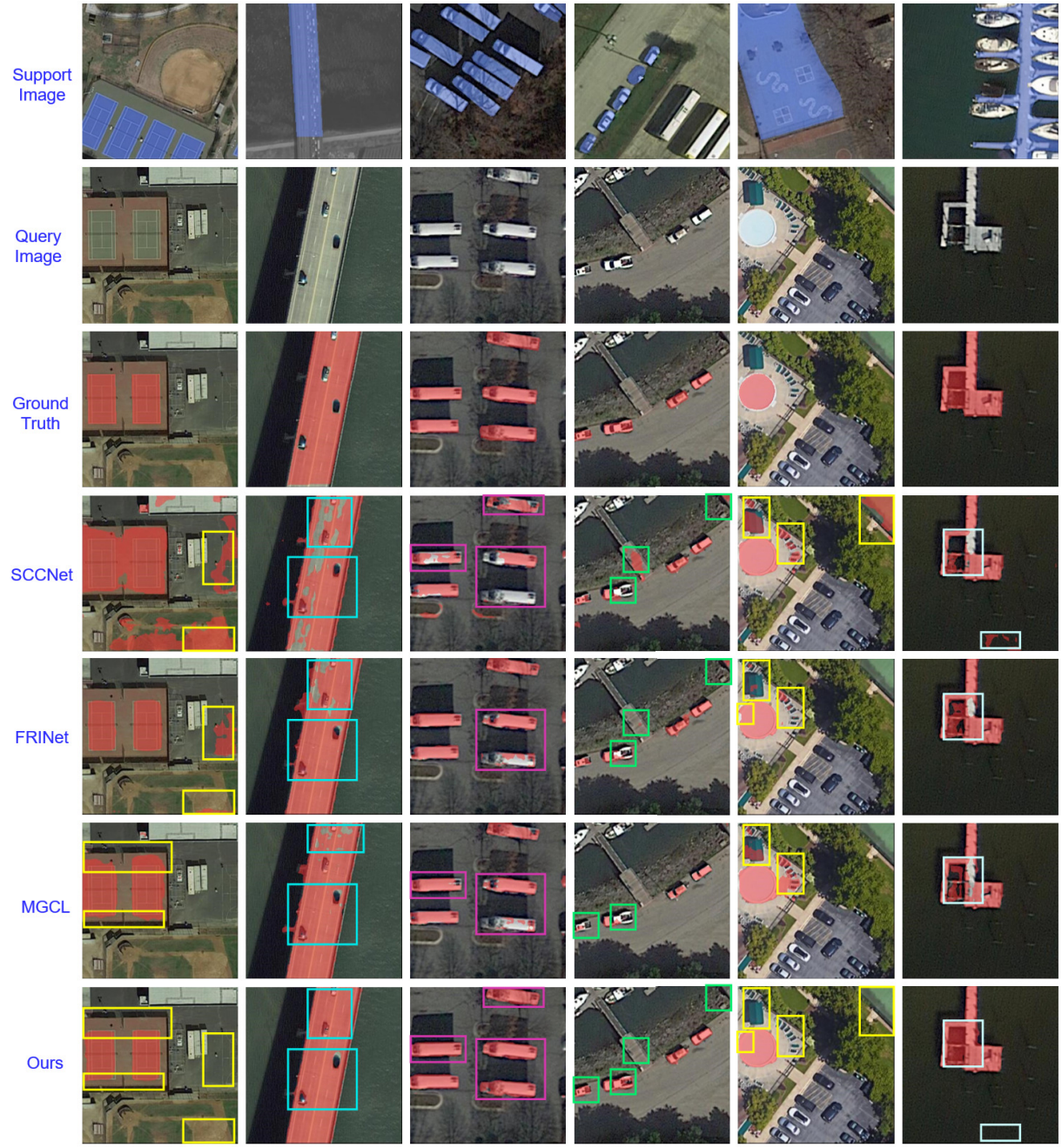


Fig. 10. Qualitative comparisons between our network and the state-of-the-art method.

object features, reducing information loss and enhancing global context modeling. Second, PRM generates customized query-guided support features, producing more discriminative prototypes robust to appearance variations. Third, hierarchical integration of MGRM and PRM creates a synergistic architecture that effectively fuses category-specific prototypes and multiscale guidance, driving overall segmentation gains.

F. Model Efficiency

The parameters of an FSS network and the inference time per episode are two crucial metrics for evaluating the model's efficiency. Using ResNet-50 as the backbone, we select five top-performing FSS models to compare their efficiency in the one-shot task.

TABLE VII
COMPARISON OF DIFFERENT MODELS IN LEARNED PARAMETERS, PER-EPISODE INFERENCE TIME, AND MIOU

Method	Parameters(M)	Inference time(ms)	mIoU(%)
SDM [35]	19.60	34.86	30.72
HDMNet [57]	**4.20**	25.40	34.34
SCCNet [67]	5.20	23.10	39.45
FRINet [36]	43.93	27.10	42.59
TBRNet (Ours)	13.30	**21.86**	**43.98**

Table VII presents the experimental results on the iSAID-5^i dataset.

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1) TBRNet requires fewer learnable parameters than SDM and FRINet, indicating superior computational efficiency.
2) Although HDMNet and SCCNet have fewer parameters, their inference time exceeds that of TBRNet, meaning that they are slower in prediction.
3) TBRNet achieves the highest inference speed among all methods, making it highly practical for real-time applications.
4) TBRNet also attains the optimal mIoU, demonstrating strong segmentation performance
5) In summary, TBRNet balances segmentation accuracy and computational complexity effectively: while its parameter count is not the lowest, it offers the best tradeoff between performance and efficiency, positioning it as a robust choice for RS tasks.

To further assess practical applicability, we calculate TBRNet's floating point operations (FLOPs), which amount to 15.2 G. This indicates that TBRNet can run smoothly on devices with moderate computational power, expanding its utility in real-world RS operations.

V. Conclusion

In this article, we present a TBRNet for FSS in RSIs. Most existing FSS methods primarily focus on natural images and face considerable challenges with RSIs due to unique attributes such as substantial intraclass variances and multicategory coexistence. To address these challenges, we design the TBRNet architecture, integrating two key modules: MGRM and PRM. Specifically, MGRM can generate multilevel guidance maps to facilitate target category identification, while PRM enhances prototype representation to mitigate the intraclass variations. Finally, a hierarchical decoder fuses these maps, enhanced prototypes, and query features to boost segmentation performance. Experimental results on two benchmark datasets validate TBRNet's superior performance over existing FSS algorithms.

Although TBRNet achieves state-of-the-art performance in RS FSS, it has limitations. Like most existing work, it focuses on novel class segmentation but may struggle in real-world scenarios where novel and base classes coexist—a key challenge in generalized FSS (GFSS). To tackle this, we plan to optimize TBRNet for GFSS, enabling unified segmentation of both class types. This will not only resolve practical limitations but also advance the field by expanding few-shot methods to more complex RS tasks.

Meanwhile, we recognize the need for dedicated datasets to support such advancements. In the future, we aim to explore diverse RS datasets and collaborate with domain experts to construct a benchmark specially tailored for FSS. This benchmark will include predefined annotation protocols (e.g., systematic base-novel class partitioning and fixed sample size configurations) to standardize dataset structures for few-shot learning, facilitating rigorous comparisons and reproducible research.

Looking further ahead, we intend to extend our methodology to zero-shot semantic segmentation in RS. By fine-tuning the architecture and introducing a module that maps learned features to novel class semantics, we aim to enable the model to segment classes with no prior training samples. This transition from few-shot to zero-shot capabilities will enhance the model's generalization and contribute to making RS analysis more autonomous and adaptive to evolving real-world demands.

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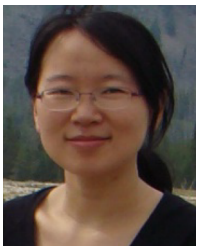
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